Mateo del Rio Lanse

US Citizen | mdelriolanse@gmail.com
Portfolio: mdelriolanse.com | LinkedIn: mdelriolanse | GitHub: mdelriolanse

EDUCATION

Cornell University, College of Engineering **Ithaca, NY**
B.S. Electrical & Computer Engineering (GPA: 3.75 / 4.00) *Expected Graduation: Dec. 2028*
Relevant Coursework: CS 4414 (Systems Programming), ECE 2300 (Digital Logic and Computer Organization), MATH 2940

PROFESSIONAL EXPERIENCE

Options Technology **New York, NY**
Software Engineering Intern, PrivateMind Platform *Jun. 2026 – Aug. 2026*

- Owned and shipped an OpenAI-compatible REST Batch API end-to-end on cloud-native Kubernetes infrastructure: async GPU workers, exactly-once billing for consistency, DB schema, SDK, Helm deployment, ~7,000 lines of tests.
- Benchmarked frontier LLMs before production serving: Poisson-distributed concurrency sweeps measuring time-to-first-token, inter-token latency, and throughput at p50/p95/p99 across multi-GPU HGX deployments.
- Owned LLM inference gateway load management on NVIDIA B200/B300 GPUs: priority scheduling so batch cannot starve interactive chat, per-model streaming timeouts, and token-budget admission control on GPU memory pressure.
- Deployed a novel text-to-image model to production (GPU config, gateway routing, bounded concurrency, pricing, UI) and built the content-safety guardrail layer: a moderation API and an actionable error taxonomy for opaque refusals.
- Merged 133 PRs across 10 repos in Rust, TypeScript, Python, SQL, and Helm through GitHub Actions CI/CD, building distributed systems for scalability; unified two agent execution paths, deleting ~6,000 legacy lines.
- Shipped a nightly memory-consolidation engine, an automation builder driven by a backend tool-calling loop, chat streaming hardened against state bugs, and an inline co-editing canvas with conflict auto-resolution.

Cornell University **Ithaca, NY**
AI Alignment Undergraduate Researcher, Cornell NLP Group *Oct. 2025 – May 2026*

- Built the AI security evaluation pipeline behind Open Weight, Open Risk (co-author): 15 frontier open-weight LLMs across 800 WMDP-derived adversarial bio/chem requests, run locally on B200-class GPUs with vLLM and SGLang.
- Co-designed the training-free inception attack: a small uncensored 7B architect model pivots target reasoning, lifting compliance from ~10% to >95% in 5 iterations, exposing shallow alignment in open-weight reasoning models.

Engineering Manager & ML Engineer, Generative AI at Cornell *Oct. 2025 – May 2026*

- Led an 8-person team on full-stack development of an ESG risk monitoring platform for Investcorp ($60B+ AUM); built a vendor due-diligence agentic data pipeline with fintech company QuickFi, reconciling vendor data from filings.

Hospital for Special Surgery **New York, NY**
Machine Learning Research Intern *Jun. 2024 – Aug. 2024*

- Analyzed immunological datasets with Dr. Amit Lakhanpal (Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, VAEs, t-SNE) to extract rheumatoid arthritis signatures; fine-tuned neural networks, generated ESM embeddings.

PROJECTS

Marko *Devpost* | *GitHub*

- Piloted an indoor AR navigation app (Swift, ARKit), mapping a multi-story Cornell academic building for routing.
- Solved indoor positioning, where GPS is unreliable, with quick, precise localization from what the camera sees and no beacons or routers, via PyTorch/RoMa and Open3D on RunPod GPUs; added an ElevenLabs voice agent.

Groundwork *Website* | *Pitch Deck*

- Co-built an on-premises AI maintenance agent that turns machinery fault signals into cited, verified work orders.
- Served Qwen3.6-35B-A3B via llama.cpp on Grace Blackwell inside OpenShell; its egress policy denied exfiltration.

Dispatch *Pitch Deck* | *GitHub*

- Built the backend of a multi-agent security platform: an orchestrator agent reads the target codebase (routes, auth logic, raw SQL), plans targeted attacks, and dispatches pentester workers into isolated Blaxel sandboxes.
- Closed the loop from finding to fix: content-hash IDs deduplicate findings, constructor workers patch code and open GitHub PRs, and an Express REST API plus Slack/Linear/GitHub integrations trigger fixes and ticket findings.

Debately *Website* | *GitHub*

- Built a moderated, unbiased debate platform with organized layouts and AI summaries.
- Engineered web-data fact-checking pipelines to prevent hallucinations and an LLM consensus tracker (Anthropic API).

AWARDS

MLH Big Red Hacks (2026) — Marko, 2nd Place
Cornell AI Hackathon (2026) — Sandboxed Agentic Penetration Testing Platform, Best Developer Tool
Dell x NVIDIA AI Hackathon at Cornell (2026) — Groundwork, 2nd Place
2x Regeneron WESEF (2024 & 2025) — 2nd Place in Computer Science

SKILLS

Programming Languages: Rust, Python, TypeScript, JavaScript, SQL, C (systems)
Technologies: Linux/Unix, Docker containers, Kubernetes, Helm, vLLM, SGLang, RESTful APIs, Git, PostgreSQL/Supabase